

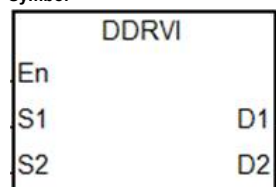
API	Instruction			Operand							Description						
2706	D	DRVI		S_1 , S_2 , D_1 , D_2							Relative Position control						

Device	X	Y	M	S	T	C	HC	D	FR	SM	SR	E	K	16#	"\$"	F
S_1							●	●	●		○		○	○		
S_2							●	●	●		○		○	○		
D_1		○														
D_2		○	○													

Data type	BOOL	WORD	DWORD	LWORD	UINT	INT	DINT	LINT	REAL	LREAL	TMR	CNT	STRING
S_1			●				●						
S_2			●				●						
D_1	●												
D_2	●												

Pulse Instruction	16-bit instruction	32-bit instruction
—	—	AS

Symbol



- S_1 : Number of output pulses (relative positioning)
- S_2 : Pulse output frequency
- D_1 : Pulse output device
- D_2 : Pulse direction output device

Explanation

- This instruction specifies the output pulse setting for relative positioning. S_1 is the number of output pulses (relative positioning). The range is between -2,147,483,648 to +2,147,483,647, and the +/- signs indicate the positive and negative pulse directions. When the instruction is executed and S_1 is 0, it indicates not outputting and output completion flag not be set to ON. If 0 is a possible output number in your program, it is suggested to add more conditions in your program to rule out this possibility.
- S_2 is the pulse output frequency, and the range is between 0 Hz–200 kHz for open collector output models and 0–4 MHz for line driver output models. If the value in S_2 is less than the Start/end frequency (Hz) set in SR (refer to the SR table below for more details), PLC operates according to the values set in SR.
- Refer to the following table for the selection of pulse output devices for D_1 and D_2 . When the output mode for D_1 and D_2 is not the default Pulse+direction output (0 is the default value), modify the mode by setting SR to 1, which changes the mode into A/B phase output.

Axis number	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
Output point for D_1	Y0.0	Y0.2	Y0.4	Y0.6	Y0.8	Y0.10
Direction output point for D_2	Y0.1	Y0.3	Y0.5	Y0.7	Y0.9	Y0.11
Output mode	SR462	SR482	SR502	SR522	SR542	SR562

DDRVI and DDRVA support the output devices for D_1 which can be the odd points as shown in the following table. Only the Pulse+direction output mode is supported. It is not recommended that you use the input points Y0.0, Y0.2–Y0.10 as the high-speed output points.

Axis number	Axis 7	Axis 8	Axis 9	Axis 10	Axis 11	Axis 12
Output point for D_1	Y0.1	Y0.3	Y0.5	Y0.7	Y0.9	Y0.11
Direction output point for D_2	Y1.0 or above or any M device (BOOL)					

- The output point for the pulse direction output device D_2 is affected by the scan cycle if it is not the recommended output point in the table above.
- When D_2 uses the output mode Pulse+direction and S_1 has a minus sign, D_2 is ON. When S_1 has a plus sign, D_2 is OFF. D_2 is not OFF immediately after the pulse output in the negative direction is completed. D_2 switches to OFF when the contact for the execution of the instruction is OFF.
- For the instruction execution, you must use the parameters such as the start/end frequency and ramp-up/down time. Refer to the following table for SM/SR corresponding to the axes.

The following table applies when D_1 selects the even output points as the output devices.

Output point for D_1	Y0.0	Y0.2	Y0.4	Y0.6	Y0.8	Y0.10
Busy flag	SM460	SM480	SM500	SM520	SM540	SM560
Completion flag	SM461	SM481	SM501	SM521	SM541	SM561
Reversing the output direction	SM462	SM482	SM502	SM522	SM542	SM562
Stop flag	SM463	SM483	SM503	SM523	SM543	SM563
Enabling fixed slope ramp-up/down	SM469	SM489	SM509	SM529	SM549	SM569
Output completion auto-reset	SM470	SM490	SM510	SM530	SM550	SM570

Executing an interrupt program when pulse output ends	SM471	SM491	SM511	SM531	SM551	SM571
Corresponding interrupt I number	I500	I501	I502	I503	I504	I505
The output immediately stops when the instruction is disabled or stops	SM476	SM496	SM516	SM536	SM556	SM576
Present output position	SR460 SR461	SR480 SR481	SR500 SR501	SR520 SR521	SR540 SR541	SR560 SR561
Start/end frequency (Hz)	SR463	SR483	SR503	SR523	SR543	SR563
Ramp-up time (ms)	SR464	SR484	SR504	SR524	SR544	SR564
Ramp-down time (ms)	SR465	SR485	SR505	SR525	SR545	SR565
Target frequency of the fixed slope	SR472 SR473	SR492 SR493	SR512 SR513	SR532 SR533	SR552 SR553	SR572 SR573
Present output frequency	SR610 SR611	SR612 SR613	SR614 SR615	SR616 SR617	SR618 SR619	SR620 SR621

The following table applies when **D₁** selects the odd output points as the output devices.

Output point for D₁	Y0.1	Y0.3	Y0.5	Y0.7	Y0.9	Y0.11
Busy flag	SM472	SM492	SM512	SM532	SM552	SM572
Completion flag	SM473	SM493	SM513	SM533	SM553	SM573
Output stop flag	SM474	SM494	SM514	SM534	SM554	SM574
Output completion auto-reset	SM475	SM495	SM515	SM535	SM555	SM575
The output immediately stops when the instruction is disabled or stops	SM477	SM497	SM517	SM537	SM557	SM577
Present output position	SR474 SR475	SR494 SR495	SR514 SR515	SR534 SR535	SR554 SR555	SR574 SR575
Start/end frequency(Hz)	SR476	SR496	SR516	SR536	SR556	SR576
Ramp-up/down time (ms)	SR477	SR497	SR517	SR537	SR557	SR577

Some flags, such as fixed slope ramp-up/down, and executing an interrupt program when pulse output ends, are not supported by the odd output points.

If the output is ongoing, the preset output is the ramp-down stop when the instruction is disabled or stops temporarily. If the output must be stopped immediately, set the immediately stop flag to ON.

7. For PLC with FW V1.06.00 or later, added new SR652, SR653 for displaying number of pulse in ramp-up area and new SR654, SR655 for displaying number of pulse in ramp-down area.

Refer to the following table for Attribute, Factory default and latched values of SM/SR corresponding to the axes.

Item	Attribute	Stop→Run	Run→Stop	Power ON	Factory default	Latched
Busy flag	R	OFF	OFF	OFF	OFF	N
Completion flag	R / W	OFF	OFF	OFF	OFF	N
Stop flag	R / W	OFF	OFF	OFF	OFF	N
Output completion auto-reset	R / W	OFF	OFF	OFF	OFF	N
Enabling fixed slope ramp-up/down	R / W	OFF	OFF	OFF	OFF	N
Executing an interrupt program when pulse output ends	R / W	OFF	OFF	OFF	OFF	N
The output immediately stops when the instruction is disabled or stops	R / W	OFF	OFF	OFF	OFF	N
Present output position	R / Wd	No change	No change	No change	0	Y
Start/end frequency (Hz)	R / Wd	No change	No change	No change	200	Y
Ramp-up time (ms)	R / Wd	No change	No change	No change	200	Y
Ramp-down time (ms)	R / Wd	No change	No change	No change	200	Y
Target frequency of the fixed slope	R / Wd	No change	No change	No change	0	Y
Present output frequency	R	0	0	0	0	N

Note: R mean the device is readable. W means the device is writable. Wd means the device is writable any time except when performing the high-speed output.

8. After output completion auto-reset is enabled, the flag is automatically reset to OFF when the output completes. You must set the flag to ON again if the auto-reset function is needed for the next output operation. It is suggested that you use the rising-edge or falling-edge to trigger the flag every time. The function is usually used when the output DDRVI instruction cannot be scanned and executed by PLC program after DDRVI is executed. For example, when the instruction is executed in an interrupt program.

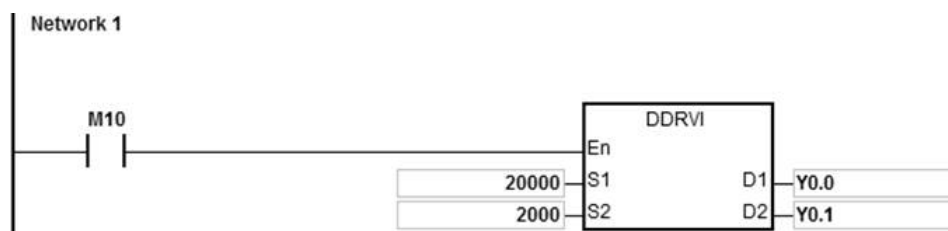
9. If you use the Completion flag to set the I interrupt function, the flag is automatically reset to OFF when the output completes and the interrupt program starts. You must set the flag to ON again for the next output in which the interrupt occurs. It is suggested that you use the rising-edge or falling-edge to trigger the flag every time.
10. When the direction outputs are not using the default outputs to output, you can refer to the table below for setting up the time for directional outputting goes first before pulse outputting. So that you can be sure the outputting only happens when switching to the direction outputs. The unit for SR is millisecond and the range is 0-20 ms. Default is 0, indicating inactive.

Output point for D_1	Y0.0	Y0.1	Y0.2	Y0.3	Y0.4	Y0.5
SR number	SR640	SR641	SR642	SR643	SR644	SR645
Output point for D_1	Y0.6	Y0.7	Y0.8	Y0.9	Y0.10	Y0.11
SR number	SR646	SR647	SR648	SR649	SR650	SR651

Note: the time source here is from PLC scan time. (available for firmware version 1.02.20 or later)

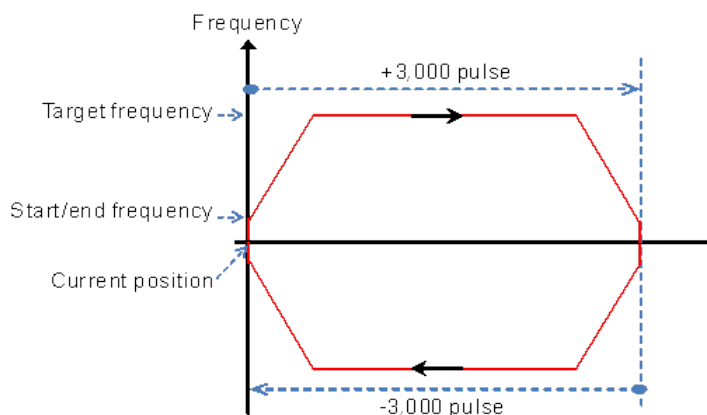
Example 1

When M10 is ON, Y0.0 outputs 20,000 pulses at 2 kHz (relative positioning). Y0.1 = OFF means the positive direction.



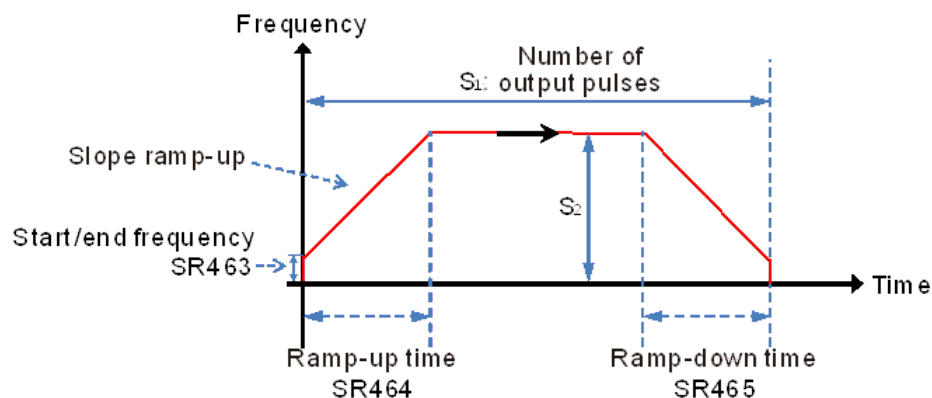
Additional remarks

1. The following graph explains the relative positioning by specifying the travel distance from the current position with the +/- signs.

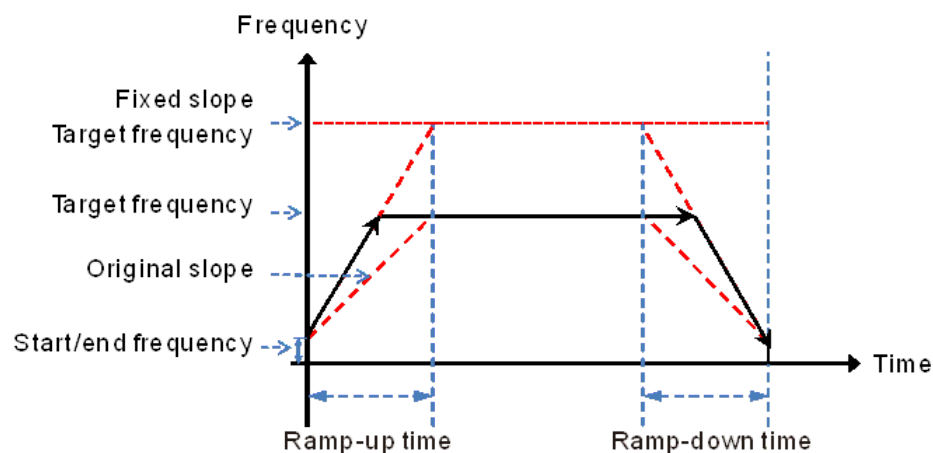


2. Setting the ramp up/down and the items for relative positioning

a) Y0.0 output curve diagram



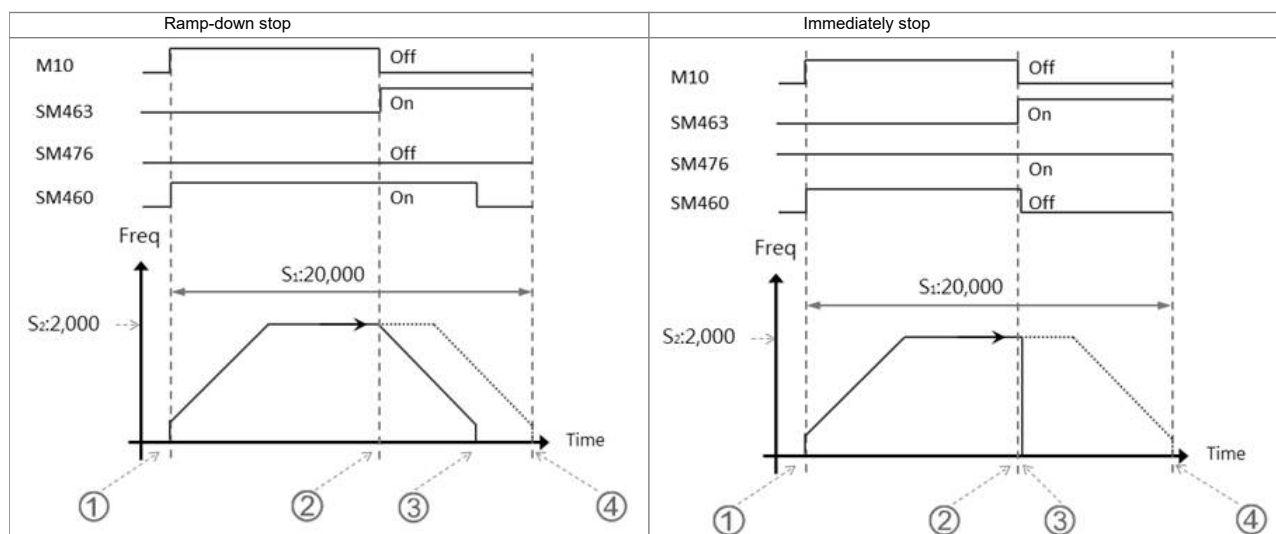
- b) You can create several DRVI instructions using the same output point in the program. The PLC can start only one DRVI instruction when executing the program. For example, when one instruction starts Y0.0 for output, then other instructions that also use Y0.0 for output are not executed. The instruction that starts the output point first uses the point first.
- c) After the instruction is executed, any modification of the parameters is not accepted until the instruction is disabled.
3. Ramp-up and ramp-down for fixed slopes
- a) The following graph shows the Y0.0 output curve diagram.



- Start the fixed slope flag SM469 and write the target frequency of the fixed slope into SR472.
- The new slope takes the place of the original slope and the DDRVI instruction performs the positioning as shown above.
- The general slope is determined by the start/end frequency, target frequency, and ramp up and down time.
The fixed slope is determined by the start/end frequency, target frequency of the fixed slope, and ramp up and down time.
- When the target frequency of the fixed slope is less than the target frequency, the fixed slope function is not started.

4. Stop flag and immediately stop flag

a) The following graphic shows the output diagram based on example 1.

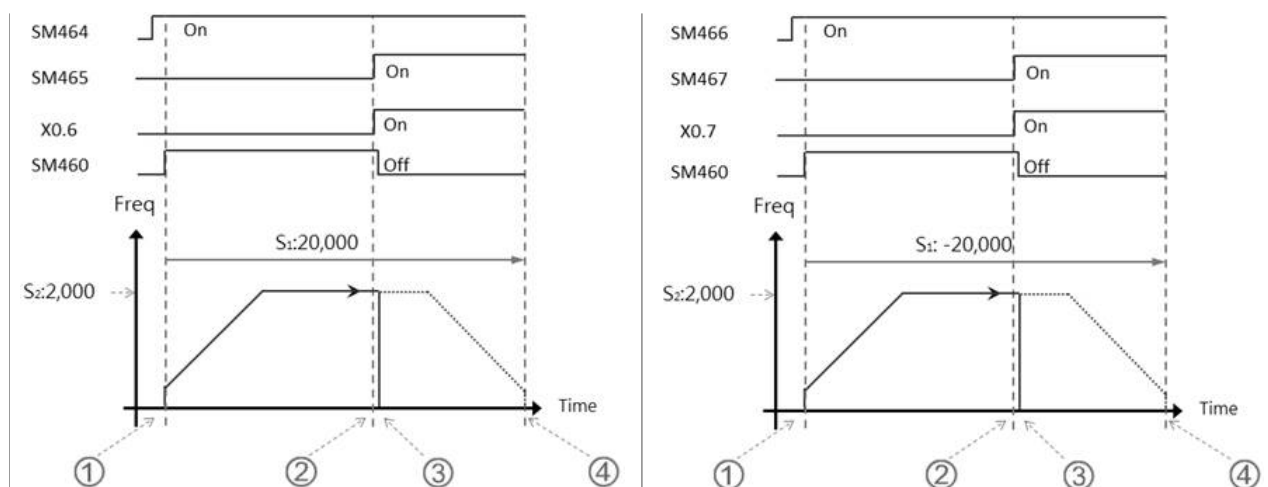


- Ramp-down stop**
 - When M10 is ON, the DDRVI instruction is executed and starts to output pulses, and SM460, the Busy flag is ON.
 - When the instruction is disabled (M10 is OFF) or the stop flag SM463 is ON, the state of the immediately stop flag SM476 is confirmed.
 - If the immediately stop flag SM476 is OFF, the output ramps down to stop and then the Busy flag SM460 is OFF.
 - If M10 is not OFF and the stop flag SM463 is not ON after the instruction is executed, Y0.0 stops the output after 20,000 pulses.
- Immediately stop**
 - When M10 is ON, the DDRVI instruction is executed and starts to output pulses, and SM460, the Busy flag is ON.
 - If the instruction is disabled (M10 is OFF) or the stop flag SM463 is ON, the state of the immediately stop flag SM476 is confirmed.
 - If the immediately stop flag SM476 is ON, the output stops immediately and the Busy flag SM460 is OFF.

5. Hardware limit function

a) The following graphic shows the output curve diagram of axis 1 (Y0.0/Y0.1).

Positive limit function	Negative limit function



b) Operation of the positive limit

① Positive limit input point X0.6 is set in HWCONFIG and downloaded to the PLC. When SM464 is set to ON, the positive limit function starts.

② DDRVI outputs 20,000 pulses in the positive direction. **Note:** the instruction does not output in the positive direction if the positive limit point is ON when the function starts.

③ When the limit input point X0.6 is triggered by the external mechanism and is ON, the condition for the hardware positive limit function is met.

④ DDRVI instruction stops the output immediately and the positive limit alarm SM465 is ON.

⑤ If the condition for the hardware limit function is not met, Y0.0 stops the output after 20,000 pulses.

c) Operation of the negative limit

① Negative limit input point X0.7 is set in HWCONFIG and is downloaded to PLC. When SM466 is set to ON, the negative limit function starts.

② DDRVI outputs 20,000 pulses in the negative direction. **Note:** the instruction does not output in the negative direction if the negative limit point is ON when the function starts.

③ When the limit input point X0.7 is triggered by the external mechanism and is ON, the condition for the hardware negative limit function is met.

④ DDRVI instruction stops the output immediately and the negative limit alarm SM467 is ON.

⑤ If the condition for the hardware limit function is not met, Y0.0 stops the output after -20,000 pulses.

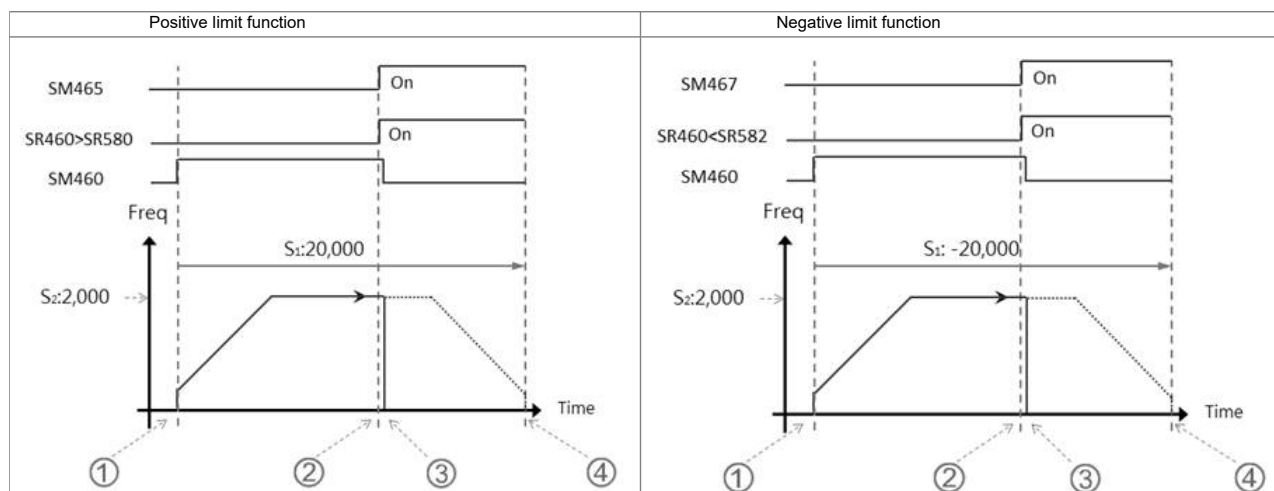
d) SM flags related to the limit function

Axis number	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
Enabling the hardware positive limit	SM464	SM484	SM504	SM524	SM544	SM564
Positive limit alarm flag	SM465	SM485	SM505	SM525	SM545	SM565
Enabling the hardware negative limit	SM466	SM486	SM506	SM526	SM546	SM566
Negative limit alarm flag	SM467	SM487	SM507	SM527	SM547	SM567

Note: The limit point alarm flag is set or reset automatically by the PLC according to the state of the start flag.

6. Software limit function

a) The following diagram shows the output curve diagram of axis 1 (Y0.0/Y0.1).



b) Operation of the positive limit

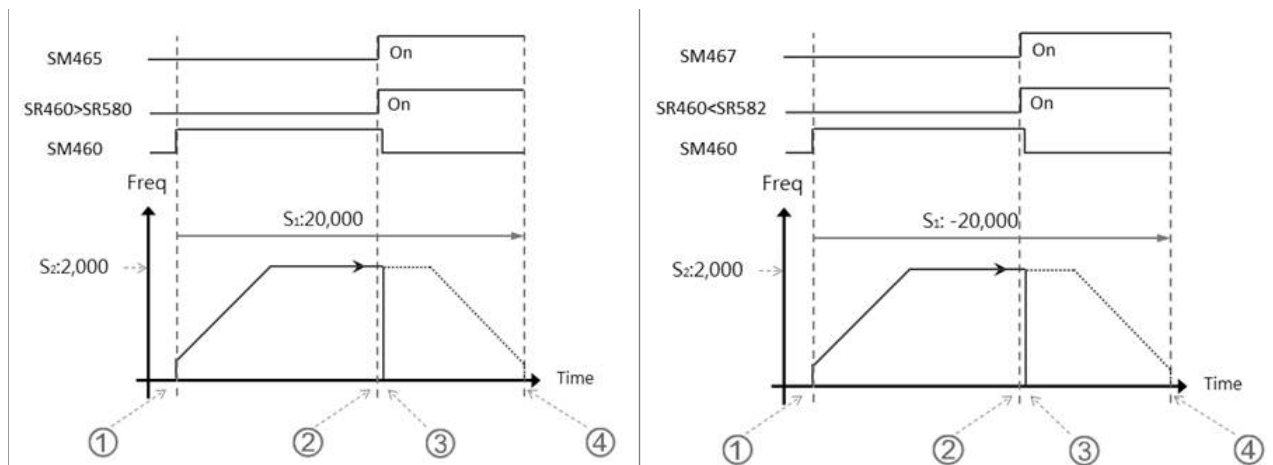
① Positive software limit position is set in HWCONFIG (by setting SR580=11,000 for example). After downloading the setting to the PLC, it means the positive limit function starts.

② DDRVI starts to output 20,000 pulses in the positive direction. **Note:** the instruction does not output in the positive direction if the positive limit is exceeded when the function starts.

③ When the current position SR460 > the limit position SR580, it means that the condition for the software limit function is met.

④ DDRVI instruction stops the output immediately and the positive limit alarm SM465 is ON.

⑤ If the condition for the limit function is not met, Y0.0 stops the output after 20,000 pulses.



c) Operation of the negative limit

① Negative software limit position is set in HWCONFIG (by setting SR582 = -11,000 for example). After downloading the setting to the PLC, the negative limit function starts.

② DDRVI starts to output 20,000 pulses in the negative direction. **Note:** the instruction does not output in the negative direction if the negative limit is exceeded when the function starts.

③ When the current position SR460 < the limit position SR582, the condition for the software negative limit function is met.

④ DDRVI instruction stops the output immediately and the negative limit alarm SM467 is ON.

⑤ If the condition for the software limit function is not met, Y0.0 stops the output after -20,000 pulses.

d) Rising-edge trigger

Limit function Types	Contact A Rising-edge trigger	Contact B Falling-edgetrigger
Descriptions	When started: from low to high and maintains at high	When started: from high to low and maintains at low

e) SR related to the limit function

Axis number	Axis 1	Axis 2	Axis 3	Axis 4	Axis 5	Axis 6
Positive software limit position	SR580 SR581	SR584 SR585	SR588 SR589	SR592 SR593	SR596 SR597	SR600 SR601
Negative software limit position	SR582 SR583	SR586 SR587	SR590 SR591	SR594 SR595	SR598 SR599	SR602 SR603

Positive output limit: the output stops immediately if the current position is greater than the positive limit position.

Negative output limit: the output stops immediately if the current position is less than the negative limit position.

When the positive and negative limits are both 0, the software limit function cannot start. Since the software limit function checks the output position when the output instruction is scanned, stopping the output is affected by the PLC scan. To quickly stop the output in real time, use the external input point as the limit point.

7. ST Example

```

0001 IF M0 THEN
0002     DDRVI(1000,1000,Y0.0,Y0.1);
0003     M1 := TRUE;
0004 ELSIF M1 AND SM461 THEN
0005     SM470 := TRUE;
0006     M1 := FALSE;
0007 END_IF;

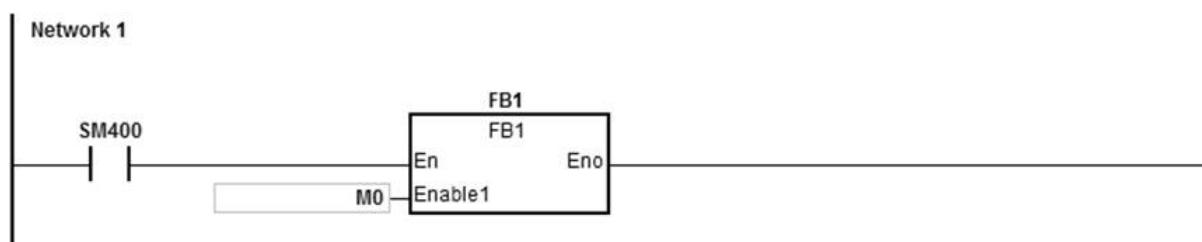
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Explanation

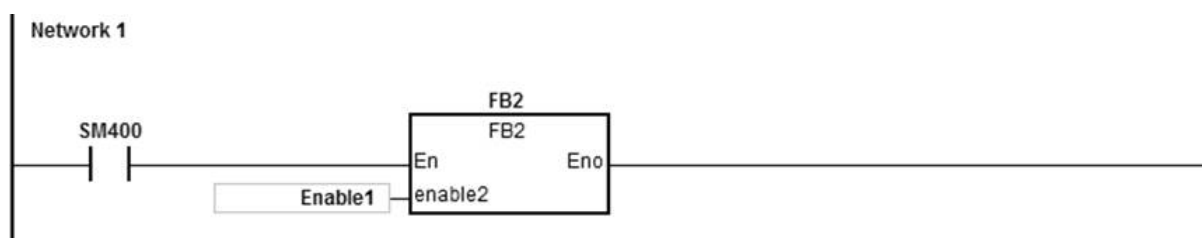
- When M0 is ON, Y0.0 outputs 1000 pulses at 1 kHz.
- When the pulse output completes, SM461 is ON and SM470=ON is triggered.
- The pulse output restarts when M0 changes from OFF to ON again
- When there are more than two high-speed output instructions in a program, do not use the variable name, M1, repeatedly.
- It is suggested not to use ST language in a program if there is any chance the outputting needed to be stopped.

8. Function block example

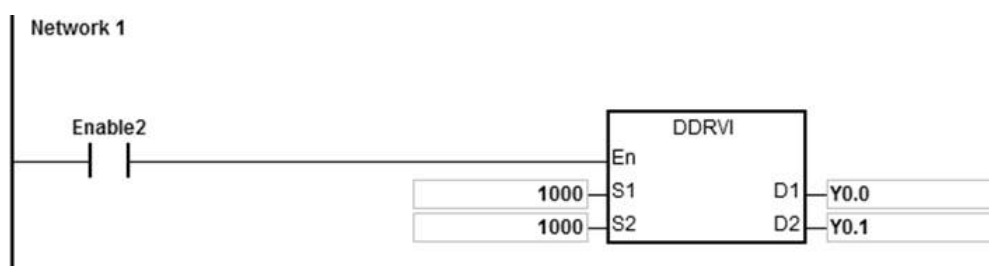
Main program



Function block 1



Function block 2



Explanation

When M0 is ON, Y0.0 outputs 10000 pulse at 1kHz. Y0.1 = OFF means the pulse output direction is positive.

- Be sure to refer to PLSR instruction (API2701) for more details on changing target frequency during outputting. Remember if there is any change on the target frequency, the change can only be carried out after the original target frequency completes outputting.
- Changing the target position during outputting:
 - This function is only available for firmware version 1.02.30 or later. You cannot change outputting direction even after the target position is changed. For example, if the original output target position is in the positive direction, after the target position is changed, it still outputs in the positive direction.
 - After you set the change flag, the changing happens when the instruction is being scanned. And after it is changed, the change flag is cleared off. But if the target position is set incorrectly, the change flag keeps ON.

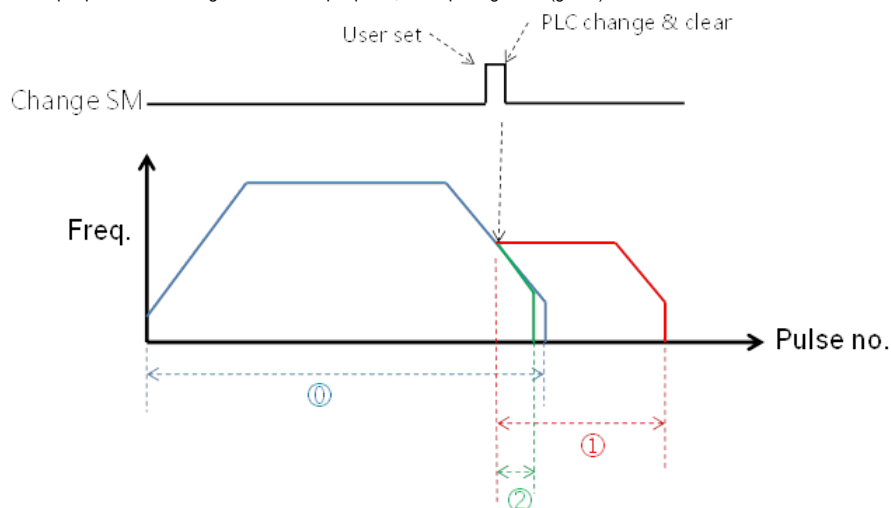
- c) Refer to the following table for the corresponding outputs and change flag SM.

Output point	Y0.0	Y0.1	Y0.2	Y0.3	Y0.4	Y0.5
SM number	SM478	SM479	SM498	SM499	SM518	SM519
Output point	Y0.6	Y0.7	Y0.8	Y0.9	Y0.10	Y0.11
SM number	SM538	SM539	SM558	SM559	SM578	SM579

- d) This changing target position function is available for DDRVI and DDRVA instructions. When the instruction DDRVI is used and a change flag is set, the change will be executed. Even if the output number is 0, the operation will be carried out and stop outputting. When the instruction DDRVA is used, if the changed target position is the same as the previous one, this change will not be performed.
- e) Three situations may occur, when changing target position during output.

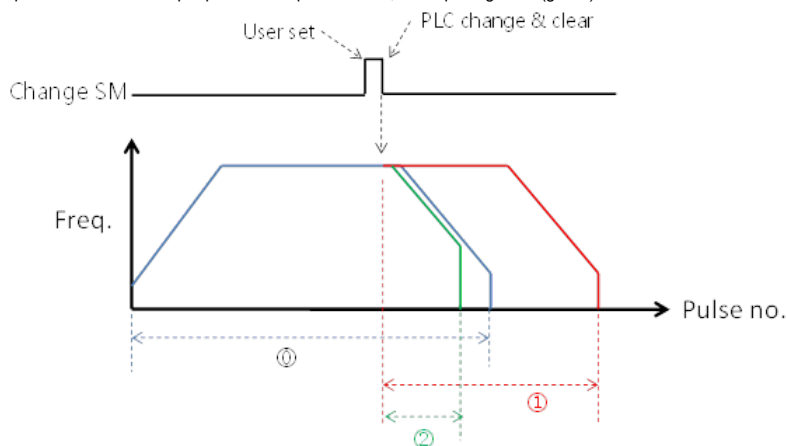
Situation A: changing timing occurs during outputting in the ramp-down area.

- ① original target position when started (blue)
- ② new number of output pulse $\geq$ remaining number of output pulse, its outputting curve (red).
- ③ new number of output pulse $<$ remaining number of output pulse, its outputting curve (green).



Situation B: changing timing occurs during outputting in the target frequency area.

- ① original target position during initial start (blue)
- ② new number of output pulse $\geq$ number of output pulse in ramp-down area, its outputting curve (red).
- ③ new number of output pulse $<$ number of output pulse in ramp-down area, its outputting curve (green).



Situation C: changing timing occurs during outputting in the ramp-up area.

- ① original target position during initial start (blue)
- ② new number of output pulse $\geq$ number of output pulse in ramp-down area, its outputting curve (red).
- ③ new number of output pulse $<$ number of output pulse in ramp-down area, its outputting curve (green).

